

UB? In My Lexer?

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Abstract

The mere act of Lexing c++ can result in undefined behavior. This paper removes that undefined behavior. Further work will be needed to remove all undefined behavior in [cpp].

Revisions

Revision 2

- Apply CWG feedback from Issaquah

Revision 1

- Fix typos

Motivation

According to the standard, the following examples expose undefined behavior:

```
int \\ // UB : universal character name accross spliced lines
```

```
u\
```

```
0\
```

```
3\
```

```
9\
```

```
1 = 0;
```

```
#define CONCAT(x, y) x ## y
```

```
int CONCAT(\\, u0393) = 0; // UB: universal character name formed by macro expansion
```

```
// UB: unterminated string
```

```
const char * foo = "
```

It does seem unfortunate that lexing C++ would incur UB. As such, we propose to change the specification to remove the UB by either well-defining the behavior or making it ill-formed, closely matching implementations. The status-quo, as well as the proposed changes, are

summarized below. The red cell highlights the only impact this paper would have on existing implementations.

	GCC	CLANG	EDG	MSVC	Proposed
Spliced UCN	Supported	Supported	Supported	Error	Well-formed
UCN Produced BY ##	Supported	Supported	Supported	Supported	Well-formed
Unterminated " or '	ill-formed	ill-formed	ill-formed	ill-formed	ill-formed

We propose that spliced UCNs be supported because, in addition to 3/4 of surveyed compilers supporting it, it falls off naturally of the specification: splicing happens before any other form of tokenization and supporting it avoid special-casing this oddity.

Wording

❖ Phases of translation [lex.phases]

2. Each sequence of a backslash character (\) immediately followed by zero or more whitespace characters other than new-line followed by a new-line character is deleted, splicing physical source lines to form logical source lines. Only the last backslash on any physical source line shall be eligible for being part of such a splice. ~~Except for splices reverted in a raw string literal, if a splice results in a character sequence that matches the syntax of a universal-character-name, the behavior is undefined.~~

[Note: Line splicing can form a *universal-character-name* [lex.charset]. — end note]

A source file that is not empty and that does not end in a new-line character, or that ends in a splice, shall be processed as if an additional new-line character were appended to the file.

A preprocessing token is the minimal lexical element of the language in translation phases 3 through 6. In this document, glyphs are used to identify elements of the basic character set [lex.charset]. The categories of preprocessing token are: header names, placeholder tokens produced by preprocessing `import` and `module` directives (*import-keyword*, *module-keyword*, and *export-keyword*), identifiers, preprocessing numbers, character literals (including user-defined character literals), string literals (including user-defined string literals), preprocessing operators and punctuators, and single non-whitespace characters that do not lexically match the other preprocessing token categories. If a U+0027 APOSTROPHE or a U+0022 QUOTATION MARK matches the last category, the ~~behavior is undefined~~ program is ill-formed. If any character not in the basic character set matches the last category, the program is ill-formed.

❖ The ## operator [cpp.concat]

A ## preprocessing token shall not occur at the beginning or at the end of a replacement list for either form of macro definition.

If, in the replacement list of a function-like macro, a parameter is immediately preceded or followed by a `##` preprocessing token, the parameter is replaced by the corresponding argument's preprocessing token sequence; however, if an argument consists of no preprocessing tokens, the parameter is replaced by a placemaker preprocessing token instead. For both object-like and function-like macro invocations, before the replacement list is reexamined for more macro names to replace, each instance of a `##` preprocessing token in the replacement list (not from an argument) is deleted and the preceding preprocessing token is concatenated with the following preprocessing token. Placemaker preprocessing tokens are handled specially: concatenation of two placemarkers results in a single placemaker preprocessing token, and concatenation of a placemaker with a non-placemaker preprocessing token results in the non-placemaker preprocessing token. *If the result begins with a sequence matching the syntax of `universal-character-name`, the behavior is undefined. [Note: This determination does not consider the replacement of `universal-character-name`s in translation phase 3[lex.phases]. —end note]* *[Note: Concatenation can form a `universal-character-name`. —end note]* If the result is not a valid preprocessing token, the behavior is undefined. The resulting token is available for further macro replacement. The order of evaluation of `##`

Future work

The reader will have noticed that `[cpp]` has a few other undefined behaviors. This should equally be fixed, however, this work is best left to someone with greater preprocessor expertise.